

The Language of Medicine

13th edition

Davi-Allen Chabner

Chapter 2

Terms Pertaining to the Body as a Whole

Chapter Goals (Slide 1 of 2)

- Define terms that apply to the structural organization of the body.
- Identify the body cavities and recognize the organs contained within those cavities.
- Locate and identify the anatomical and clinical divisions of the abdomen.
- Locate and name the anatomical divisions of the back.

Chapter Goals (Slide 2 of 2)

- Become acquainted with terms that describe positions, directions, and planes of the body.
- Identify the meanings for new word elements and use them to understand medical terms.

Chapter 2

Lesson 2.1

Structural Organization and Body Cavities

- Define terms that apply to the structural organization of the body.
- Identify the body cavities and recognize the organs contained within those cavities.
- Locate and identify the anatomic and clinical divisions of the abdomen

Structural Organization of the Body

(Slide 1 of 7)

- **CELLS**
- **TISSUES**
- **ORGANS**
- **SYSTEMS**

Structural Organization of the Body

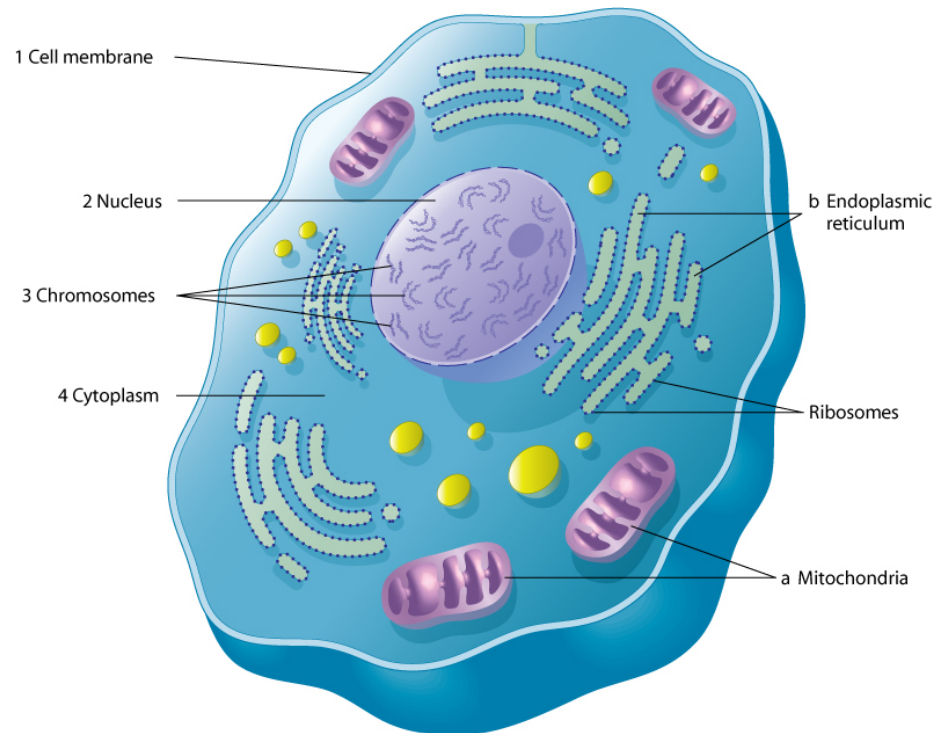
(Slide 2 of 7)

Cells—The fundamental unit of all living things

- Major parts of a cell include:
 - Chromosomes
 - Nucleus
 - Cell membrane
 - Cytoplasm
 - Mitochondria
 - Endoplasmic reticulum

Structural Organization of the Body (Slide 3 of 7)

- **CELL—The fundamental unit of all living things**



Study Section 1

- Anabolism
- Catabolism
- Cell membrane
- Chromosomes
- Cytoplasm
- DNA
- Endoplasmic reticulum
- Genes
- Karyotype
- Metabolism
- Mitochondria
- Nucleus

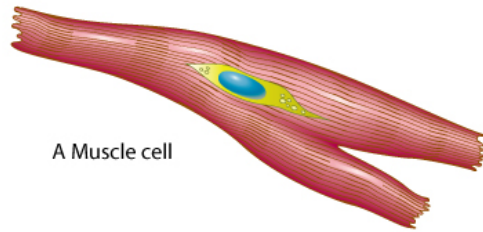
Structural Organization of the Body

(Slide 4 of 7)

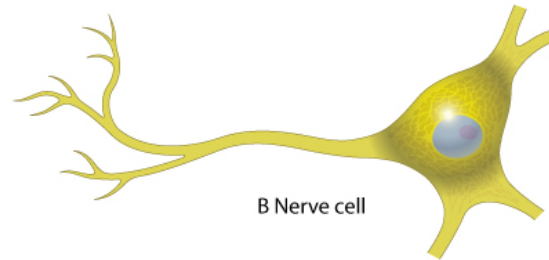
- Cells are specialized to carry out individual functions.
- Types of cells include:
 - Muscle cell
 - Nerve cell
 - Epithelial cell
 - Fat cell

Structural Organization of the Body (Slide 5 of 7)

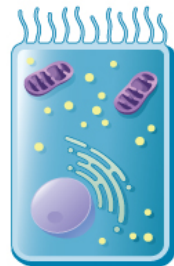
- **Types of cells**



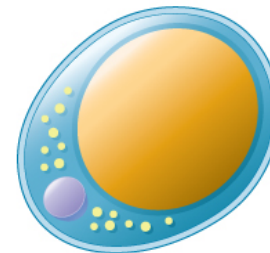
A Muscle cell



B Nerve cell



C Epithelial cell



D Fat cell

Structural Organization of the Body

(Slide 6 of 7)

Tissues

- A tissue is a group of similar cells working together to do a specific job.

Organs

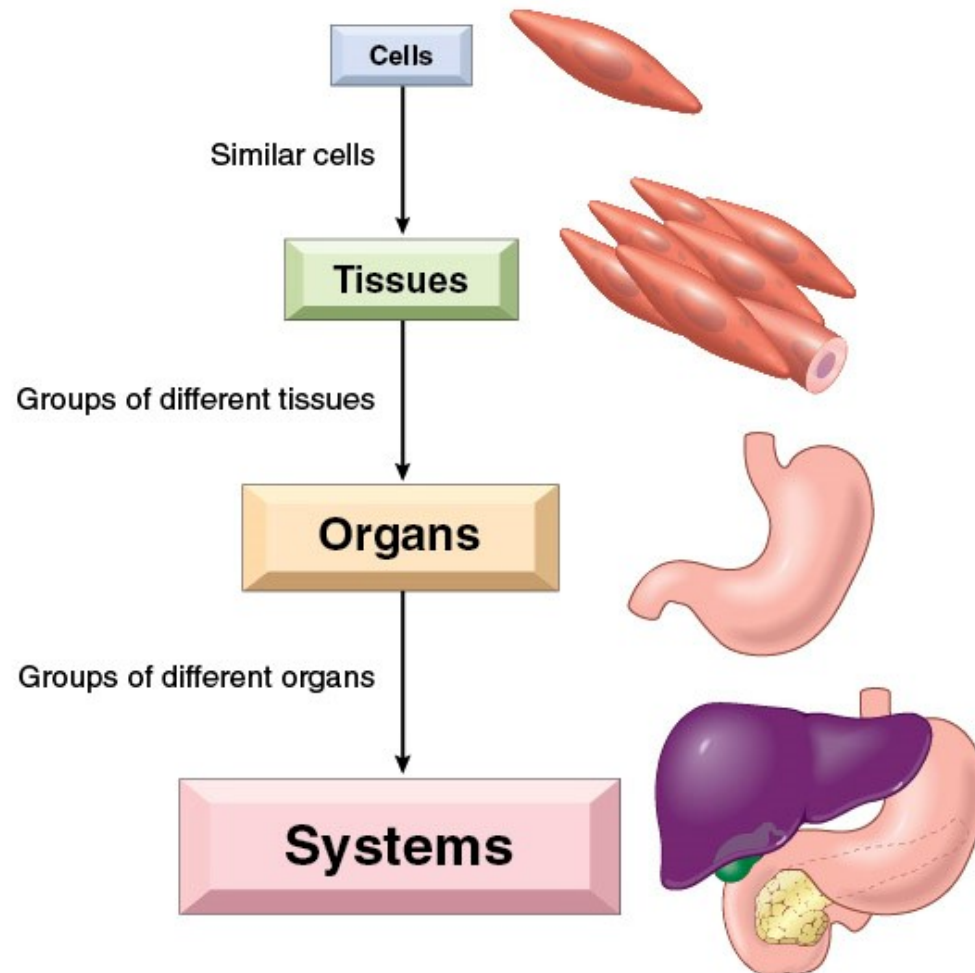
- Different types of tissue combine to form an organ

Systems

- Groups of organs working together to perform complex functions

Structural Organization of the Body

(Slide 7 of 7)



Study Section 2

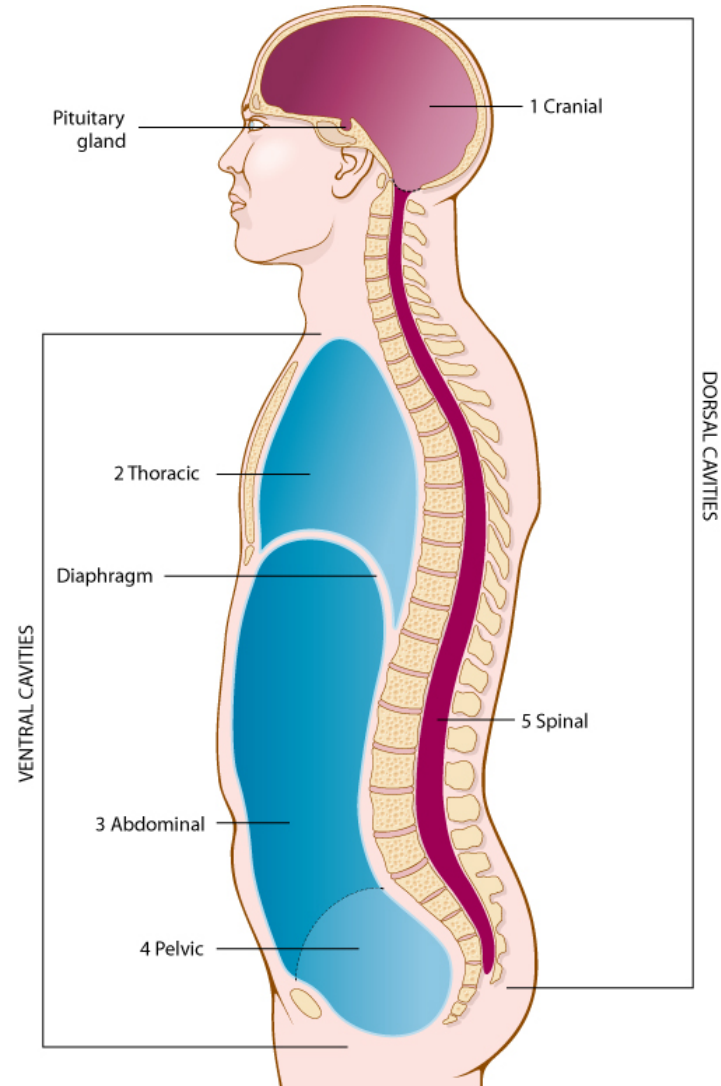
- Adipose tissue
- Cartilage
- Epithelial cells
- Histologist
- Larynx
- Pharynx
- Pituitary gland
- Thyroid gland
- Trachea
- Ureter
- Urethra
- Uterus
- Viscera

Body Cavities (Slide 1 of 2)

- Body cavities are the spaces within the body that contain internal organs (viscera) and include:
 - Cranial
 - Thoracic
 - Abdominal Pelvic
 - Spinal

Body Cavities (Slide 2 of 2)

- **Body cavities are the spaces within the body that contain internal organs (viscera).**



Body Cavities and Organs

Cavity	Organ(s)
Cranial	Brain
Thoracic	Lungs, heart, esophagus, trachea, bronchial tubes, thymus gland, aorta
Abdominal	Stomach, small and large intestines, spleen, pancreas, liver, gall bladder
Pelvic	Portions of the small and large intestines, bladder, rectum, urethra, ureters; uterus and vagina in female
Spinal	Nerves of the spinal cord

QUICK QUIZ (Slide 1 of 5)

- 1. Which body cavity contains the pleural cavity and the mediastinum?**
 - A. Cranial
 - B. Pelvic
 - C. Spinal
 - D. Thoracic

QUICK QUIZ (Slide 2 of 5)

2. Which body cavity contains the peritoneum?

A. Abdominal

B. Pelvic

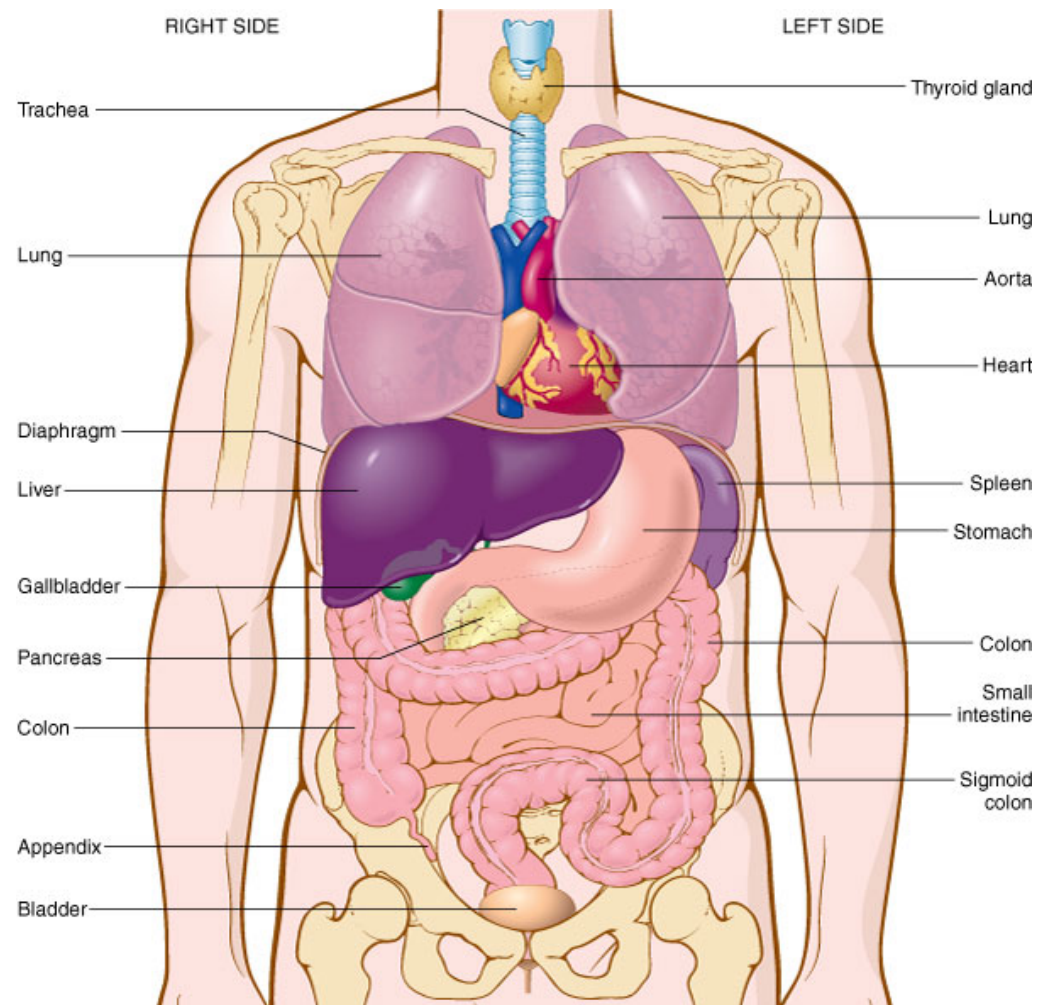
C. Spinal

D. Thoracic

Organs of the Abdominopelvic and Thoracic Cavities (Slide 1 of 2)

- Trachea
- Lung
- Diaphragm
- Liver
- Gallbladder
- Pancreas
- Colon
- Appendix
- Bladder
- Thyroid gland
- Aorta
- Heart
- Spleen
- Stomach
- Colon
- Small intestine
- Sigmoid colon

Organs of the Abdominopelvic and Thoracic Cavities (Slide 2 of 2)



Study Section 3

- Abdominal cavity
- Cranial cavity
- Diaphragm
- Dorsal (posterior)
- Mediastinum
- Pelvic cavity
- Peritoneum
- Pleura
- Pleural cavity
- Spinal cavity
- Thoracic cavity
- Ventral (anterior)

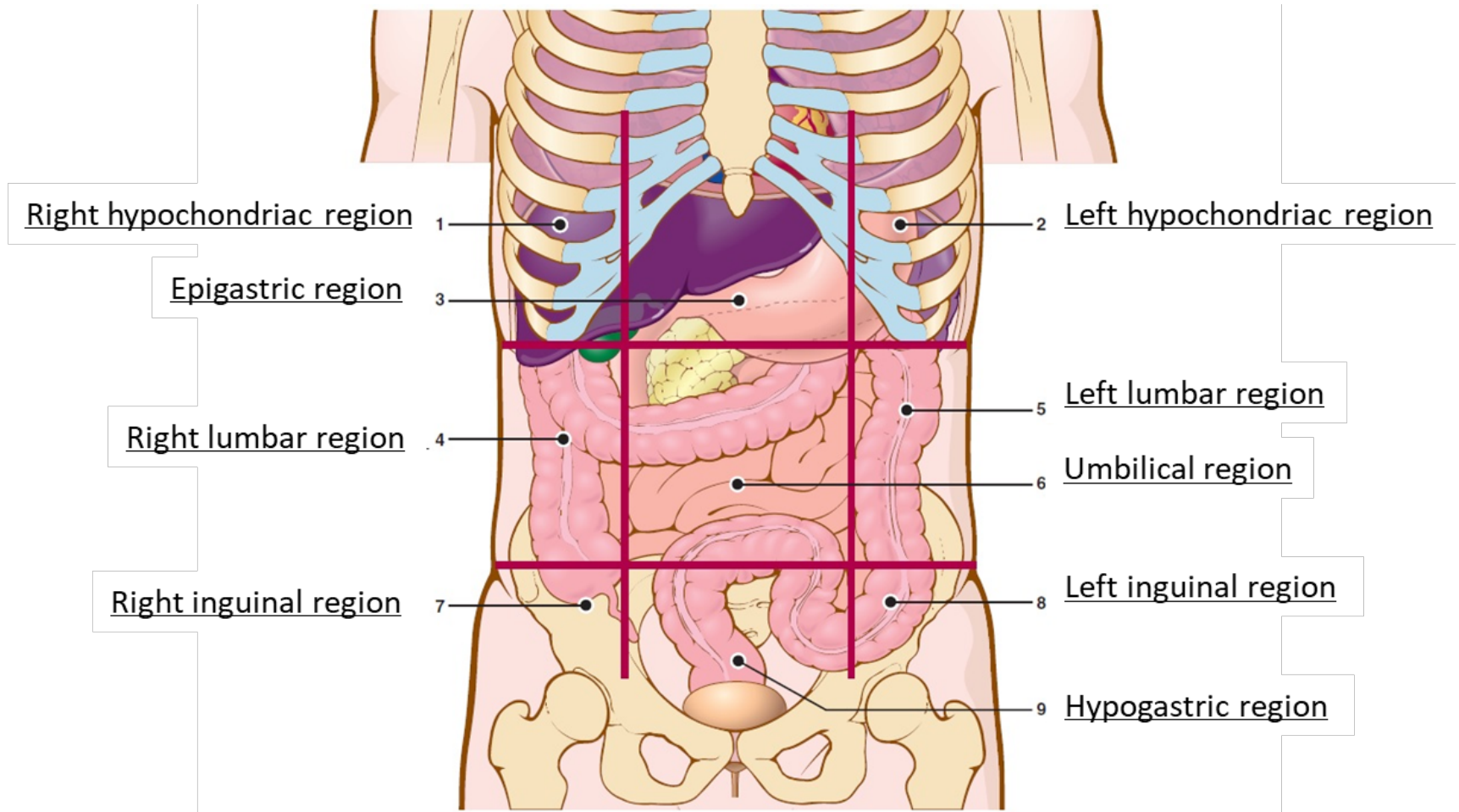
Abdominopelvic Regions

(Slide 1 of 2)

- Right hypochondriac region
- Left hypochondriac region
- Epigastric region
- Right lumbar region
- Left lumbar region
- Umbilical region
- Right inguinal region
- Left inguinal region
- Hypogastric region

Abdominopelvic Regions

(Slide 2 of 2)



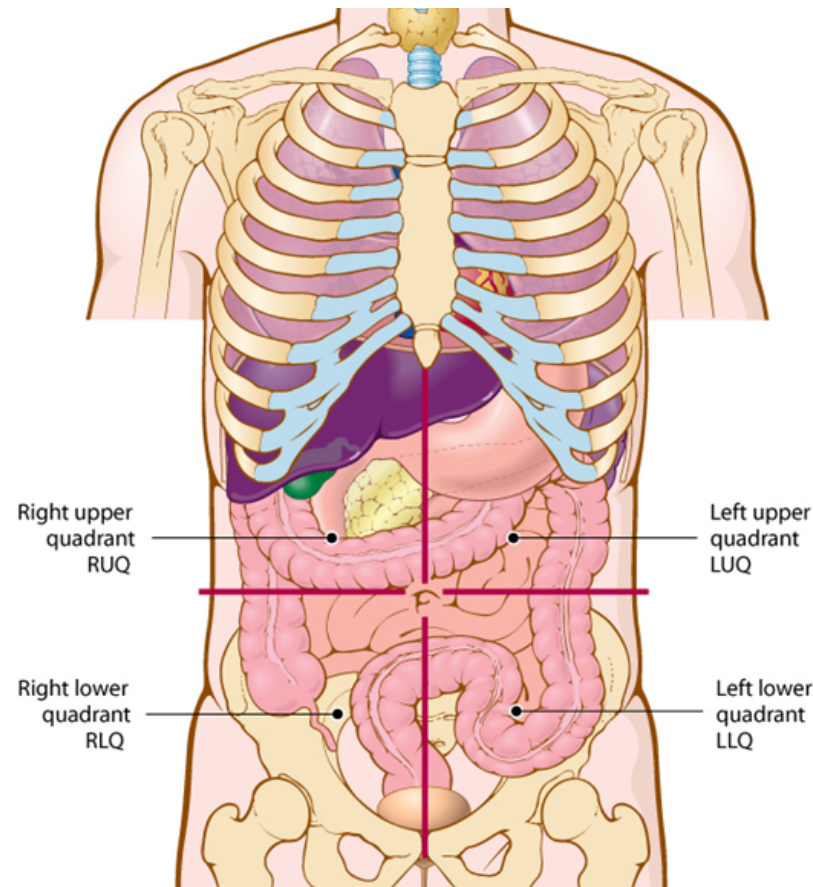
Abdominopelvic Quadrants

(Slide 1 of 2)

- **Right upper quadrant (RUQ):** contains liver (right lobe), gallbladder, part of the pancreas, parts of small and large intestines
- **Left upper quadrant (LUQ):** contains liver (left lobe), stomach, spleen, part of the pancreas, parts of small and large intestines
- **Right lower quadrant (RLQ):** contains parts of small and large intestines, right ovary, right fallopian tube, appendix, right ureter
- **Left lower quadrant (LLQ):** contains parts of small and large intestines, left ovary, left fallopian tube, left ureter

Abdominopelvic Quadrants

(Slide 2 of 2)



Chapter 2

Lesson 2.2

Positions, Directions, Planes, and Terminology

- Locate and name the anatomic divisions of the back.
- Become acquainted with terms that describe positions, directions, and planes of the body.
- Identify the meaning for new word elements and use them to understand medical terms.

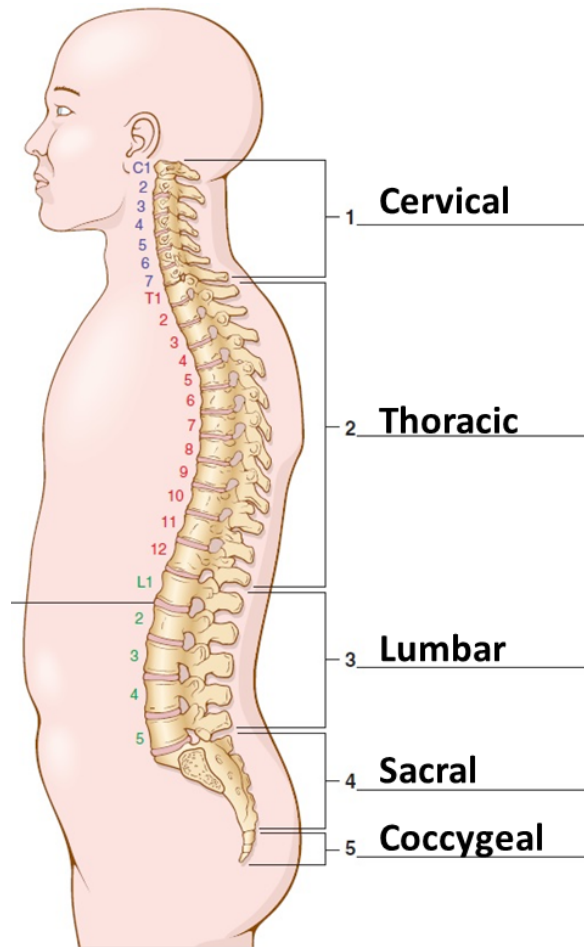
Anatomical Divisions of the Back

(Slide 1 of 2)

- Cervical
- Thoracic
- Lumbar
- Sacral
- Coccygeal

Anatomical Divisions of the Back

(Slide 2 of 2)



Study Section 4 (Slide 1 of 2)

- **Abdominopelvic Regions:**

- Hypochondriac
- Epigastric
- Lumbar
- Umbilical
- Inguinal
- Hypogastric

- **Abdominopelvic Quadrants:**

- RUQ
- LUQ
- RLQ
- LLQ

Study Section 4 (Slide 2 of 2)

- Divisions of the Back:
 - Cervical
 - Thoracic
 - Lumbar
 - Sacral
 - Coccygeal
- Related Terms:
 - Vertebra
 - Vertebrae
 - Spinal column
 - Spinal cord
 - Disc

QUICK QUIZ (Slide 3 of 5)

- 3. Name the abdominopelvic region that is located in the central region near the navel.**
- A. Inguinal
 - B. Umbilical
 - C. Lumbar
 - D. Hypochondriac

QUICK QUIZ (Slide 4 of 5)

4. Name the abdominopelvic region that is located in the lower middle below the umbilical region.

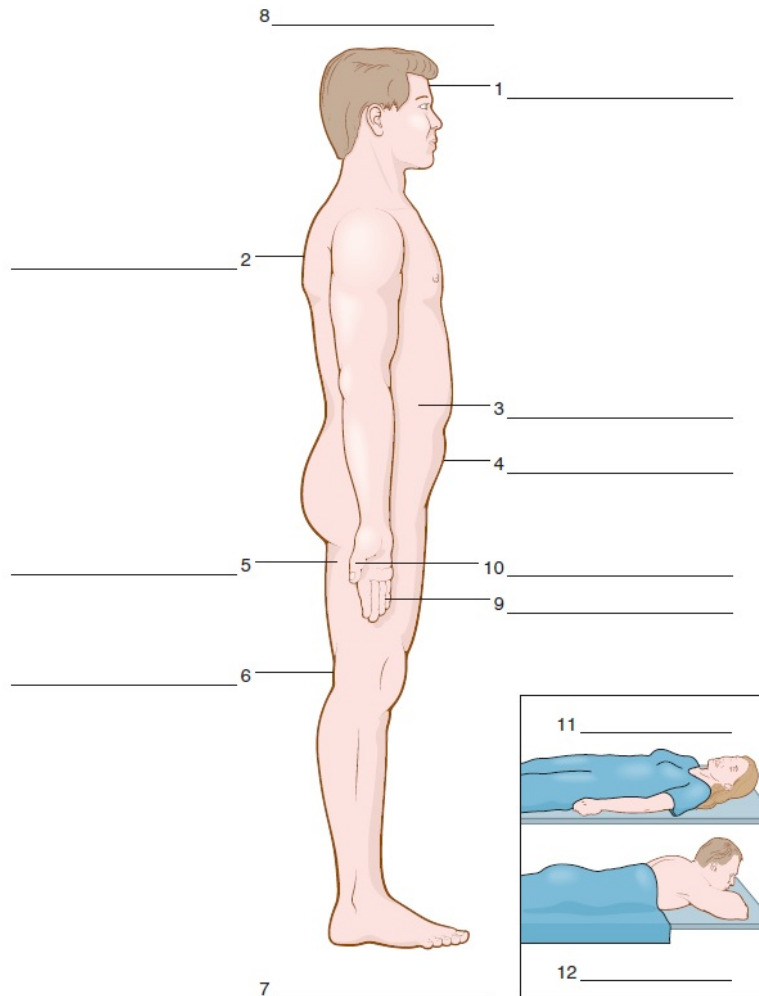
- A. Inguinal
- B. Epigastric
- C. Lumbar
- D. Hypogastric

Positional and Directional Terms of the Body

Positional and Directional Terms

- Anterior
- Posterior
- Deep
- Superficial
- Proximal
- Distal
- Inferior
- Superior
- Medial
- Lateral
- Supine
- Prone

Positional and Directional Terms

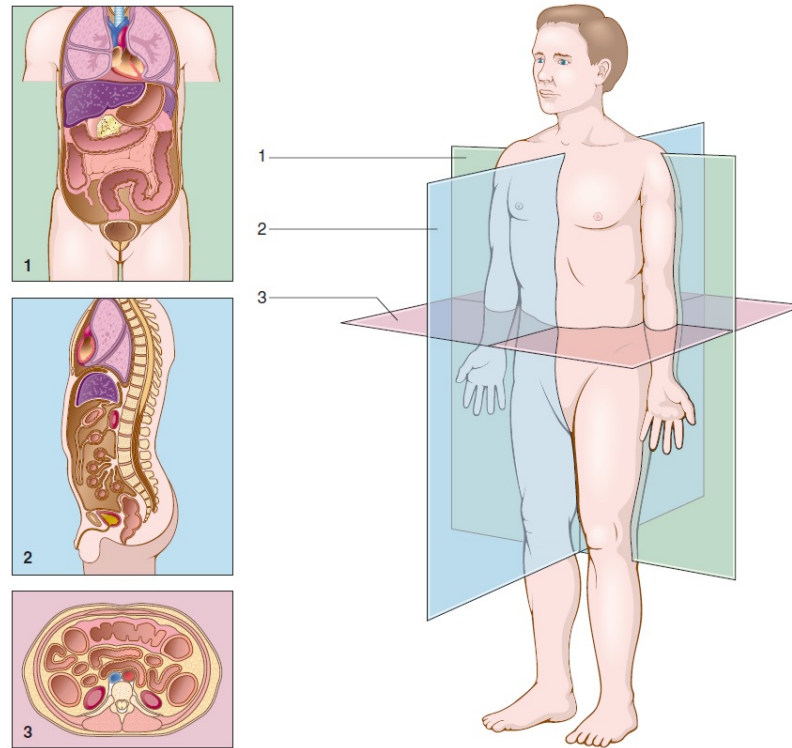


- ☐ 1. anterior
- ☐ 2. posterior
- ☐ 3. deep
- ☐ 4. superficial
- ☐ 5. proximal
- ☐ 6. distal
- ☐ 7. inferior
- ☐ 8. superior
- ☐ 9. medial
- ☐ 10. lateral
- ☐ 11. supine
- ☐ 12. prone

Planes of the Body (Slide 1 of 2)

- Frontal (coronal)
- Sagittal (lateral)
- Transverse (cross-sectional or axial)

Planes of the Body (Slide 2 of 2)



1. Frontal (coronal) 2. Sagittal (lateral) 3. Transverse (axial)

Study Section 5

- Anterior (ventral)
- Deep
- Distal
- Frontal (coronal) plane
- Inferior
- Lateral
- Medial
- Posterior (dorsal)
- Prone
- Proximal
- Sagittal (lateral) plane
- Superficial
- Superior (cephalic)
- Supine
- Transverse (axial) plane

Combining Forms, Suffixes, and Prefixes (Slide 1 of 20)

Combining Form

abdomin/o

adip/o

anter/o

Combining Forms, Suffixes, and Prefixes

(Slide 2 of 20)

Combining Form	Meaning
abdomin/o	abdomen
adip/o	fat
anter/o	front

Combining Forms, Suffixes, and Prefixes

(Slide 3 of 20)

Combining Form

cervic/o

chondr/o

chrom/o

coccyg/o

Combining Forms, Suffixes, and Prefixes

(Slide 4 of 20)

Combining Form	Meaning
cervic/o	neck (of the body or uterus)
chondr/o	cartilage
chrom/o	color
coccyg/o	coccyx (tailbone)

Combining Forms, Suffixes, and Prefixes

(Slide 5 of 20)

Combining Form

crani/o

cyt/o

dist/o

dors/o

Combining Forms, Suffixes, and Prefixes

(Slide 6 of 20)

Combining Form	Meaning
crani/o	skull
cyt/o	cell
dist/o	far, distant
dors/o	back portion of the body

Combining Forms, Suffixes, and Prefixes

(Slide 7 of 20)

Combining Form

hist/o, histi/o

ili/o

inguin/o

kary/o

Combining Forms, Suffixes, and Prefixes

(Slide 8 of 20)

Combining Form	Meaning
hist/o, histi/o	tissue
ili/o	ilium (of the hip bone)
inguin/o	groin
kary/o	nucleus

Combining Forms, Suffixes, and Prefixes (Slide 9 of 20)

Combining Form

later/o

lumb/o

medi/o

nucle/o

Combining Forms, Suffixes, and Prefixes

(Slide 10 of 20)

Combining Form	Meaning
later/o	side
lumb/o	lower back
medi/o	middle
nucle/o	nucleus

Combining Forms, Suffixes, and Prefixes (Slide 11 of 20)

Combining Form

pelv/i

pharyng/o

poster/o

proxim/o

sacr/o

Combining Forms, Suffixes, and Prefixes

(Slide 12 of 20)

Combining Form	Meaning
pelv/i	pelvis
pharyng/o	pharynx (throat)
poster/o	back, behind
proxim/o	nearest
sacr/o	sacrum

Combining Forms, Suffixes, and Prefixes (Slide 13 of 20)

Combining Form

sarc/o

spin/o

thel/o, theli/o

thorac/o

trache/o

Combining Forms, Suffixes, and Prefixes

(Slide 14 of 20)

Combining Form	Meaning
sarc/o	flesh
spin/o	spine, backbone
thel/o, theli/o	nipple
thorac/o	chest
trache/o	trachea, windpipe

Combining Forms, Suffixes, and Prefixes

(Slide 15 of 20)

Combining Form

umbilic/o

ventr/o

vertebr/o

viscer/o

Combining Forms, Suffixes, and Prefixes

(Slide 16 of 20)

Combining Form	Meaning
umbilic/o	navel, umbilicus
ventr/o	belly side of the body
vertebr/o	vertebra(e), backbone(s)
viscer/o	internal organs

Combining Forms, Suffixes, and Prefixes (Slide 17 of 20)

Prefix

ana-

cata-

epi-

hypo-

inter-

intra-

meta-

Combining Forms, Suffixes, and Prefixes

(Slide 18 of 20)

Prefix	Meaning
ana-	up
cata-	down
epi-	above
hypo-	below
inter-	between
intra-	within
meta-	change

Combining Forms, Suffixes, and Prefixes (Slide 19 of 20)

Suffix

-eal

-iac

-ior

-ism

-ose

-plasm

-somes

-type

Combining Forms, Suffixes, and Prefixes

(Slide 20 of 20)

Suffix	Meaning
-eal	pertaining to
-iac	pertaining to
-ior	pertaining to
-ism	process, condition
-ose	pertaining to, full of
-plasm	formation
-somes	bodies
-type	picture, classification

QUICK QUIZ (Slide 5 of 5)

5. Which term means process of cutting the skull?

A. Thoracotomy

B. Craniotomy

C. Discectomy

D. Arthroscopy